

Supermarket Transactions Data Analysis Report

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Introduction

I worked on a project where I analyzed supermarket transaction data provided by Commonwealth Bank in the file `supermarket_transactions.xlsx`.

The main objective of this project was to answer business-related questions about customer purchases, payment methods, and spending behavior across supermarket locations in Australia.

I completed the analysis using Python in Google Colab with the help of the Pandas library. I also used spreadsheet concepts to verify the results and better understand the dataset.

Tools and Technologies Used

For this project, I used the following tools and technologies:

- Python
- Pandas Library
- Google Colab
- Microsoft Excel

These tools helped me load, filter, analyze, and summarize the transaction data efficiently.

Understanding the Dataset

Before starting the analysis, I explored the dataset to understand its structure and identify the columns required for the task.

The dataset included information such as:

- Product names
- Quantity of products purchased
- Payment methods
- Store locations
- Customer membership types
- Total transaction amounts

To understand the data better, I used Pandas functions such as:

- `head()` — to view the first few rows of the dataset
- `columns` — to check the available column names

This helped me identify the fields needed to answer the business questions.

Analysis Performed

Question 1

How many apples were bought with cash across all locations?

To answer this question, I filtered the dataset to include only the transactions where:

- The product name was “Apple”
- The payment method was “Cash”

After filtering the data, I used the `sum()` function on the quantity column to calculate the total number of apples purchased with cash.

Result

I found that 117 apples were bought using cash payments.

Question 2

How much cash was spent on these apples?

Using the filtered dataset from Question 1, I calculated the total amount spent on apples purchased using cash payments.

I used the `sum()` function to add all transaction amounts together.

Result

The total cash spent on apples was \$537.03.

Question 3

How much money did non-member customers spend at the Bakershire store location using all payment methods?

For this analysis, I filtered the dataset to include only transactions where:

- The store location was “Bakershire”
- The customer type was “Non-member”

After filtering the records, I calculated the total spending by summing the transaction amounts.

Result

I found that non-member customers spent a total of \$2857.51 at the Bakershire store location.

Final Results Summary

Analysis Question	Final Result
Apples bought with cash	117
Total cash spent on apples	\$537.03
Bakershire non-member customer spending	\$2857.51

Conclusion

I successfully analyzed the supermarket transaction data provided by Commonwealth Bank using Python and Pandas.

During this project, I used data exploration, filtering, and aggregation techniques to answer business questions and generate useful insights from the dataset.

This project helped me improve my understanding of:

- Working with real-world transaction data
- Filtering data using Pandas
- Using aggregation functions like `sum()`
- Solving business problems using Python
- Reporting and presenting analytical findings

Overall, this project gave me practical experience in performing business-focused data analysis in a professional workflow.